

SRIVARDHAN BABA VEMULA

Buffalo, NY — srivardhanbaba@gmail.com — +1-716-604-4929 — Portfolio — LinkedIn — GitHub — Tableau

Professional Summary

Cloud & Data Systems Engineer with **3+ years** of experience architecting high-throughput ETL/ELT pipelines, fault-tolerant streaming infrastructure, and production MLOps workflows. Proven record of delivering **40% throughput gains**, optimizing large-scale distributed workloads, and maintaining **99.8% infrastructure uptime**. Driving systems development for deepfake detection platforms under **Prof. Siwei Lyu** (IEEE Fellow, Director — UB Institute for AI & Data Science).

Technical Skills

Languages	Python (Advanced), SQL, Java, R, Bash, Unix Shell Scripting
Data Engineering	ETL/ELT, Data Modeling (Star/Snowflake Schemas), Data Lakes, Delta Lake, Parquet
Cloud & Distributed Systems	AWS (S3, EC2, Lambda, CloudWatch), Azure (ADF, Azure SQL), PySpark, Apache Kafka, Hadoop
MLOps & AI Infrastructure	MLflow, Flask, Production Model Serving, REST APIs, TensorFlow, scikit-learn
Databases & Warehouses	PostgreSQL, Snowflake, Google BigQuery, MySQL, Microsoft SQL Server
DevOps & Orchestration	Apache Airflow (DAG Design, SLA Monitoring), CI/CD Pipelines, Git, Docker, SFG
Data Quality & BI	Automated Validation (Pytest/Great Expectations), Tableau, Power BI, Excel Power Query

Education

University at Buffalo — M.S. Data Science

Aug 2025 – Dec 2026

Coursework: Machine Learning, Statistical Learning, Data Mining, Predictive Analytics, Probability, Database Systems, SQL, Numerical Computing

Experience

Graduate Research Assistant — AI & Data Systems

Jan 2026 – Present

University at Buffalo | UB Media Forensic Lab | Advisor: Prof. Siwei Lyu, IEEE Fellow, AI & DS Dept. Director

- Architected **Python ETL pipelines** loading **50K+ multi-modal records** into an **AWS S3 data lake**; slashing preprocessing errors by **45%** for **Deep-O-Meter** (zinc.cse.buffalo.edu).
- Engineered **PySpark distributed workflows** for feature extraction on **100GB+ datasets**; halved computation time by **60%** via adaptive partitioning and broadcast joins.
- Designed **ML feature pipelines** (PCA, SMOTE, normalization) for **5+ deepfake models**; lifting F1-score by **12%** through structured training data.
- Built **Flask REST APIs** for production ML inference and model serving; lowered latency from **580ms to 378ms** and improved throughput by **28%**.
- Established **MLflow model versioning, experiment tracking, and model lifecycle management** across **10+ configurations**; streamlined QA effort by **30%**.

Cloud & Data Engineering Associate

Sep 2024 – Jul 2025

Broadridge Financial Solutions, India

- Automated **AWS S3 + SFTP ingestion workflows** processing **200+ files daily**; boosted throughput by **40%** and eliminated **15 hrs/week** of manual reconciliation.
- Integrated **Sterling File Gateway** with **5+ enterprise systems**; enforced SOX-compliant data lineage, lowering cross-system errors by **35%**.
- Constructed **SLA observability frameworks** (CloudWatch + PagerDuty) for **12+ pipelines**, sustaining **99.8% uptime**; trimmed incident resolution by **25%** via Airflow retry logic.
- Refactored **20+ SQL queries** to CTEs and window functions; restructured **4 schemas**, shrinking query latency by **30%** and storage overhead by **20%**.
- Instituted **CI/CD test suites** across **15+ workflows**, cutting post-deployment defects by **40%**.

Data Analyst & Python Developer

Aug 2022 – Aug 2024

Freelance, Remote

- Productionized **Python ingestion pipelines** processing **500K+ records** from **10+ sources**; reduced execution time by **45%** via vectorized transforms.
- Restructured relational schemas with composite indexing, achieving **60% faster queries** across **8 datasets** for 3 enterprise clients.
- Shipped **15+ Tableau and Power BI dashboards** saving clients **6+ hrs/week** in manual reporting.
- Maintained data workflows with validation and exception handling, sustaining **99.5% accuracy** over **2M+ records**.

Projects

- Real-Time Streaming Platform** (*Kafka, Spark Streaming, AWS S3*): Achieved **sub-2s latency** on **500K+ events/hour**; sustained **99.9% uptime** at 3× traffic spikes.
- Distributed Data Processing** (*PySpark, Hadoop*): Processed **100GB+ datasets**, halving batch runtime by **65%** via broadcast joins and adaptive partitioning.
- Azure Cloud Data Warehouse** (*ADF, Snowflake, SQL*): Unified **6 sources** into Snowflake transforming **50M+ records/month**; shortened load SLA from **4 hrs to 45 min**.
- Customer Analytics & Churn Prediction** (*scikit-learn, XGBoost*): Trained classifiers achieving **85% accuracy** via feature engineering and cross-validation.
- Workflow Orchestration Platform** (*Airflow, PostgreSQL*): Authored **10+ DAGs** eliminating **20+ hrs/week** manual scheduling; reached **98% SLA compliance**.

Certifications

- Cloud Platforms:** AWS Essentials for Developers · Azure Fundamentals (AZ-900) · Azure AI Fundamentals (AI-900)
- Data Systems & Analytics:** IBM Data Science Professional · IBM Data Analyst Professional · IBM Data Visualization Core
- Generative AI Systems:** Microsoft & LinkedIn Career Essentials in Generative AI Foundation